

Predictors of frequency risk, and how easily they are faked

a spectral-clustering attempt, a two-column physical model, and three layers of validation that reverse the answer twice

1 What is being predicted, and how it is scored

Risk is one number per operating state: the 500 ms RoCoF summed over the five worst buses and averaged over the twenty hazard scenarios (the `rocof_top5` column; the vertical axis of the right-hand pane of Figure 1). The ensemble is 240 balanced configurations of a frozen 100-bus toy grid, 16 draws at each of 15 SNSP levels from 30% to 100%.

Sections 3–6 hold out *configurations*: sixty configurations (25%, stratified by SNSP level) were held out before any modelling and read once, at the end. Everything else — the number of clusters, the feature set, the ridge penalty — was chosen by 20×5 -fold cross-validation inside the remaining 180. All models are fitted to $\log_{10}$ risk, so a residual is a multiplicative error; the “98% band” is the multiplicative width of the central 98% of residuals, and “outliers” counts held-out configurations missed by more than 10%.

That protocol is not sufficient, and Section 8 shows why. Every fold of it shares the same twenty disturbances, so it cannot detect a feature that has been built from those disturbances. The reader who wants the honest numbers should read Section 8 before believing Table 1.

2 The construction that was proposed

The grid was cut into electrically coherent regions by spectral clustering of the coupling matrix K_{ij} — the synchronising coefficients, i.e. the network as the swing equations see it. With $L = I - D^{-1/2} K D^{-1/2}$, the eigengap heuristic chooses k maximising $\lambda_{k+1} - \lambda_k$; the k lowest eigenvectors are row-normalised and k -means clustered (Ng–Jordan–Weiss). The partition depends only on the network, which never varies, so it is computed once and every configuration is described by the same regions.

This gives $k = 4$ — a clean north-west / north-east / south-east / south-west cut, wind-heavy west against demand-heavy east, cutting 11% of the total coupling. **But it wins by a hair:** the gap at $k = 4$ exceeds the runner-up ($k = 12$) by a factor of 1.01. A 10×10 mesh is not a barbell, and that margin is the first warning.

Each configuration then gets, per region, its non-synchronous share, stored energy $\sum H_i S_i$, primary reserve, mean distance to the nearest spinning machine, and the ratio (non-synchronous output) / $\sum H_i S_i$. A continuous alternative was also built: projections of the dispatch onto the ten slowest Laplacian modes (`modal`), which is what a Fiedler vector actually is before it is thresholded into a partition.

3 Results

The goal was met: against SNSP the trend narrows from $1.78\times$ to $1.18\times$ and outliers fall from 28 to 2 of 60. But the winner is not the cluster model. It is a **two-column** model containing no geography at all.

Table 1: Held-out performance on the 60 configurations never used for fitting or selection. p is the number of columns.

feature set	p	test R^2	98% band	worst miss	outliers
SNSP alone	1	0.794	1.78×	1.49×	28/60
SNSP, quadratic	2	0.833	1.75×	1.47×	23/60
$\log(\Pi/H)$, coefficients forced $(+1, -1)$	1	0.944	1.31×	1.16×	15/60
swing : $\log \Pi$, $\log H$ free	2	0.971	1.22×	1.12×	2/60
+ reserve, distance to a machine	4	0.978	1.18×	1.10×	2/60
all system-wide columns	12	0.928	1.39×	1.22×	19/60
SNSP per region	4	0.808	1.82×	1.48×	28/60
all per-region features	20	0.957	1.33×	1.18×	11/60
projection onto 10 slowest modes	31	0.971	1.25×	1.13×	5/60

4 Three controls say the clustering is not doing the work

1. **Partition control.** The identical 20-column construction on partitions that are not spectral: spectral 0.957, a naive geographic quartering drawn with a ruler 0.956, random partitions of identical sizes 0.953 on average. The spectral cut beats only **12 of 20** random partitions.
2. **Basis control.** The modal family on the ten *slowest* modes scores 0.971; on the ten *fastest* modes, 0.970. The fast modes are single-bus wiggles with no regional meaning whatever. If they predict as well as the Fiedler vector, the spectrum is not what is being used — the dimension is. Random orthonormal rotations reach 0.963.
3. **k sweep.** Held-out R^2 for the cluster families is flat and non-monotonic from $k = 2$ to $k = 12$ (0.86 to 0.95), consistent with an eigengap that had no strong preference to begin with.

5 Why there was nothing local to find

The diagnostic that explains all three controls: evaluate the same 500 ms window on the centre-of-inertia frequency — the whole grid as a single machine. That one system-wide number already accounts for $R^2 = 0.953$ of log risk. The five worst buses run 7.94× (s.d. 0.83) the system-wide reading, and only **11.8%** of the target’s variance is local amplification. The score is a system-wide quantity wearing a local costume, so a partition has almost nothing to resolve.

Does this hold for rocof_top1? Yes, but less strongly, and the trend is exactly as expected. Narrowing from five worst buses to the single worst bus raises the local share, because averaging over five buses averages local variation away:

target	R^2 vs. system-wide	local share of variance	spectral beats
rocof_top1	0.938	19.1%	15 of 20 random
rocof_top5	0.953	11.8%	12 of 20 random
rocof_top10	0.960	8.8%	—

So **rocof_top1** is genuinely more local, and the spectral partition does correspondingly better — 15 of 20 rather than 12 of 20. That is the right direction, and it is still not a result: the target remains 94% system-wide, the spectral cut still only ties the ruler-drawn quartering (+0.002), and slow and fast modes still tie exactly (0.962 versus 0.962). The conclusion survives the sharper target. Incidentally the deviation scores are *more* system-wide still (**dev_top5**: $R^2 = 0.997$,

4.8% local), which makes sense — a nadir develops over seconds, by which time the grid has synchronised, whereas a 500 ms RoCoF is read while it has not.

6 So why *do* these predictors beat SNSP?

This is the substantive point, and it has nothing to do with geography.

What the hazard variable is. First, a definition, because the symbol is easy to misread. Every configuration in this ensemble is *balanced* before anything happens: $\sum_i P_i = 4 \times 10^{-16}$ p.u. There is no standing power difference and none is being regressed on. The regressor is the aggregate *size of the contingency set*,

$$\Pi \equiv \sum_{k=1}^E |\Delta P_k|, \quad \Delta P_k = \sum_i U_{ik}, \quad (1)$$

where U_k is the injection step that disturbance k applies and $E = 20$. So ΔP_k is the imbalance each contingency *creates* at the instant it occurs — a 0.4 p.u. demand block switching in, a wind farm losing its output — not any imbalance present beforehand. Π varies between configurations because the disturbance set is *sourced*: each event scales with its own source, so a wind farm running harder has more to lose when it trips. It is known from the dispatch before any solve.

Why those two variables. For a single event the swing equation for the centre of inertia gives

$$\left. \frac{df}{dt} \right|_{t=0} = \frac{f_0 \Delta P_k}{2 \sum_i H_i S_i}, \quad (2)$$

and since the target averages over events and the response is linear in the step, the aggregate Π and the stored energy $\sum_i H_i S_i$ are the two quantities the physics says should matter. Risk is a *ratio of two extensive quantities*: megawatts that can be lost, over stored energy to absorb them. SNSP is neither. It is

$$\text{SNSP} = \frac{P_{\text{non-sync}}}{P_{\text{total}}}, \quad (3)$$

a dimensionless *share*. Three consequences follow, and together they are the entire gap in Table 1.

(i) It is a proxy for the denominator, and an imperfect one. Across the ensemble SNSP correlates with $\log \sum H_i S_i$ at $r = -0.919$. That is why SNSP works at all: displacing synchronous generation removes inertia. But it is a correlation, not an identity, because *which* machines are displaced is free. At a fixed SNSP level the stored energy still ranges by 28% of its mean (median across levels; up to 51%). Used alone, $\log(1/H)$ already scores 0.842 against SNSP's 0.794 — the proxy is worse than the thing it proxies for.

(ii) It is nearly blind to the numerator. SNSP correlates with $\log \Pi$ at only $r = +0.336$. The size of the largest credible loss depends on what is actually running and how hard — a loaded HVDC link, a radial wind farm at full output — and a *share* cannot see it. At fixed SNSP the hazard still ranges by 23% of its mean. This is the piece SNSP misses most completely, and it is why adding $\log \Pi$ to $\log H$ moves outliers from 15 to 2.

(iii) A ratio discards the scale. Two configurations with the same SNSP at different demand carry the same label and different physics. In this ensemble demand is fixed, which flatters SNSP considerably; on a real system, where demand varies by a factor of two between night and peak, the defect would be far worse.

The fitted exponents are not $(+1, -1)$, and that is itself physics. Forcing the coefficients to the purely inertial values — fitting only an offset to $\log \Pi - \log \sum H_i S_i$ — already reaches $R^2 = 0.944$. Freeing them reaches 0.971, at

$$\log_{10} \text{RoCoF} = +1.285 \log_{10} \Pi - 0.645 \log_{10} \sum_i H_i S_i + \text{const.} \quad (4)$$

The departure is not noise. The score is measured over a 500 ms window, not at $t = 0$: by then load damping and governor action have begun to arrest the frequency, and in the settled limit the offset is $\Delta P / (R + D)$, independent of inertia altogether. The inertia exponent therefore interpolates between -1 (instantaneous) and 0 (settled), and -0.645 is where 500 ms falls for this system. Consistently, the residual of the forced model correlates at $r = +0.879$ with stored energy and $r = +0.879$ with reserve — the pure inertial form over-weights inertia. The hazard exponent exceeds 1 because Π sums twenty events with different spatial signatures, so risk is proportional to a *weighted* sum of the $|\Delta P_k|$ rather than to their plain total, and the two regressors are mildly correlated ($r = -0.235$). Equation 2 is thus the right *motivation* for the two variables, not a law being confirmed to two decimal places.

What the elaborate feature sets were actually doing. Every per-region and modal feature is a *linear functional of the per-bus dispatch and inertia vectors*. Summing $H_i S_i$ over region j , or projecting it onto eigenvector v_m , both produce weighted sums of the same 100 numbers. Given twenty or thirty such functionals, a linear model can reconstruct the two aggregates in Equation 2 to good accuracy — and *any* spanning set of functionals will do, which is precisely what the partition and basis controls found. The eigenvectors are one basis among many; the fast modes are another; a random rotation is a third. They all work because they all span the same thing. Supporting this directly: the raw per-unit state (18 unit outputs plus 5 inertias plus 5 drops, no partition and no spectrum) scores 0.943, in the same band.

The clustering, in other words, was an expensive and indirect way of recovering Π and $\sum_i H_i S_i$. Computing them straight from the dispatch is two columns, needs no eigenvector, and does better.

7 Three layers of validation

Everything above holds out configurations. It never holds out *disturbances* — and Π is summed over exactly the twenty disturbances whose outcomes define the target. The predictor knows the realised sizes of the very events it is scored on, and no amount of holding out configurations can see that. What follows is what happened when the protocol was tightened twice.

Layer 1: hold out configurations (Sections 3–6)

Sixty configurations held out, one fixed twenty-event set throughout. The two-column swing model reaches $R^2 = 0.978$ with 2 outliers of 60 and looks decisive. It is not trustworthy, for the reason just given.

Layer 2: split the frozen event set in half — and why this was also wrong

Fit on ten of the twenty disturbances, rank configurations on the disjoint ten. Under this test the swing model collapsed to $\rho = 0.698$, the worst of everything tried, below plain SNSP by 0.124 ($p = 3 \times 10^{-13}$, 36 of 40 splits). Taken at face value that says Π is pure leakage.

It is not, and the test was faulty. The twenty events are a *designed* set with a fixed composition — five demand steps, eight wind drops, four unit trips. A random half of them is not a representative sample of anything: the two halves carry different mixtures, so Π_A and Π_B measure different quantities. Their aggregates *anti-correlate*, $\text{corr}(\log \Pi_A, \log \Pi_B) = -0.385 \pm 0.109$ over 200 splits, and the collapse follows from that artefact rather than from any property of the hazard.

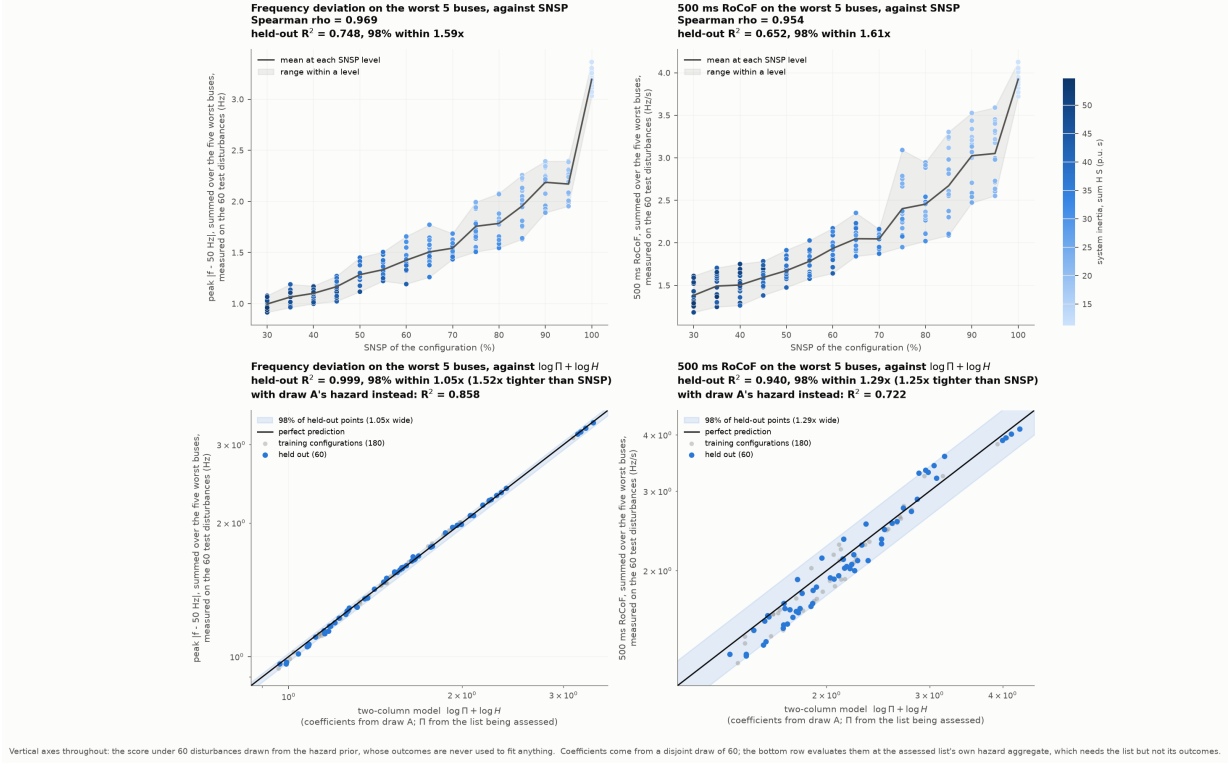


Figure 1: Vertical axes throughout: the score under draw B, sixty disturbances sampled from the hazard prior whose outcomes are never used to fit anything. Top: against SNSP. Bottom: against $\log \Pi + \log H$, two coefficients fitted on the disjoint draw A and evaluated at the assessed list's own hazard aggregate. The 98% band narrows $1.52\times$ for the deviation and $1.25\times$ for the RoCoF; each title also quotes what the same model scores holding a stale list.

Layer 3: sample two representative draws from a hazard prior

The correct version of the same question. Define a prior over disturbances of the same six families with the location and severity *drawn* rather than chosen — a demand block of 0.2 pu to 0.6 pu at any city, a front taking 30% to 90% of any farm's output, a trip of any unit capped at one machine. Draw two independent stratified samples of sixty. Both are representative of the same distribution; they contain different disturbances. Fit on draw A, predict the score under draw B.

The diagnostic reverses immediately: $\text{corr}(\log \Pi_A, \log \Pi_B) = +0.650$, and $\log \Pi_A$ correlates at $+0.899$ with the closed-form $\mathbb{E}|\Delta P|$ under the prior. Π is a property of the operating state — the expected megawatts at risk given the dispatch — estimated with sampling noise. With ten events from a designed set that noise swamped it; with sixty drawn events it does not.

Table 2: Fit on draw A, predict the score under the independent draw B. Sixty held-out configurations; bootstrap 90% intervals over them.

model	columns	test R^2	90% CI
<i>ceiling: the measured score on draw A</i>	—	<i>0.792</i>	[0.706, 0.843]
projection onto 10 slowest modes	31	0.769	[0.676, 0.824]
all per-region features	20	0.765	[0.665, 0.826]
$\log \Pi + \log H$	2	0.722	[0.595, 0.795]
every system-wide column	12	0.716	[0.575, 0.798]
inertia only	1	0.686	[0.573, 0.752]
SNSP alone	1	0.652	[0.432, 0.776]

The swing model does not collapse. It reaches 91% of the ceiling and beats SNSP by $+0.077$

in R^2 . On rank correlation the ordering differs slightly — per-region features 0.970, modal 0.962, SNSP 0.943, swing 0.932, against a ceiling of 0.964 — so the two-column model recovers the level better than the ordering.

But nothing here is resolvable. Every paired bootstrap interval straddles zero: swing − SNSP = +0.077 [−0.009, +0.186]; swing − inertia alone = +0.034 [−0.022, +0.088]; per-region − swing = +0.046 [−0.005, +0.105]. With sixty test configurations that is the resolution available. The defensible statement is that the swing model, the spatial families and SNSP all sit in one band, with swing and the spatial families probably ahead of SNSP.

Layer 3b: two questions, not one

Table 2 answers the harder of two distinct questions, and it is worth separating them because the model is far better at one than the other.

- (a) *Under our hazard prior, what is the expected risk?* Here the contingency list of the future is unknown, and the best available hazard column is $\mathbb{E}|\Delta P|$ under the prior — or a stale estimate of it from an old list. This is Table 2: $R^2 = 0.722$ for the RoCoF, 0.858 for the deviation.
- (b) *Given this contingency list, how risky is this configuration?* Here the list is known. And Π for a list is a sum of injection steps — computable from the dispatch and the list alone, with no solve. Knowing which disturbances you will be assessed against is not knowing what they will do; an operator has the list before running any study.

Evaluating the *same* coefficients, fitted on draw A, at draw B’s own hazard aggregate answers (b):

Table 3: Coefficients fitted on draw A throughout; the score is always draw B’s, whose outcomes are never used. The last column plugs draw B’s own Π into the fitted map — the list, not its results.

score	SNSP (cannot adapt)	$\log \Pi_A + \log H$ (stale list)	$\log \Pi_B + \log H$ (assessed list)
500 ms RoCoF, worst 5	0.652	0.722	0.940
deviation, worst 5	0.748	0.858	0.999

Two things follow. First, most of the gap in Table 2 was not the model failing but a *level* offset: draw B’s sampled disturbances happen to be 9.5% milder than draw A’s, so a predictor holding the old list over-predicts by a constant. SNSP has no mechanism to notice this; $\log \Pi + \log H$ corrects for it exactly, because the hazard aggregate of the new list is a quantity you can compute. That adaptability is a property SNSP structurally cannot have.

Second, $R^2 = 0.999$ on the deviation should be read with care. It is not suspicious, it is the linear response showing through: the deviation is 99.7% a system-wide quantity (Section 5), and the settled system-wide offset is $\Delta P/(R + D)$, so the score is very nearly Π divided by a damping term that this ensemble’s inertia column tracks closely ($r = 0.74$ between log reserve and log inertia). The two-column model is recovering an analytic solution. On a real system, with protection, saturation and genuine nonlinearity, it would not be this clean — the number is a property of the toy model as much as of the predictor.

Which argues for the two-column model. If the families cannot be separated on accuracy, they can be separated on how much freedom they have to be wrong. $\log \Pi + \log H$ carries two fitted coefficients; the per-region set carries twenty and the modal set thirty-one, on 180 training rows. It also states its own assumption openly: it needs a hazard *model* to compute Π , which is

exactly the contingency list an operator already maintains — and Layer 3 shows that the specific realisation of that list does not matter, only the distribution it is drawn from.

8 Conclusions

1. **The validation protocol, not the model, decided the answer — twice.** Holding out configurations alone made a leaky model look decisive; halving a designed event set made the same model look worthless. Only a hazard prior with two representative draws measures what was actually wanted.
2. **Π is a legitimate state variable.** Two independent sixty-event draws agree on it at $r = +0.650$ and it tracks the closed-form $\mathbb{E}|\Delta P|$ at $+0.899$. It is the expected megawatts at risk given the dispatch, not a memorised property of one contingency list.
3. **On a fresh draw, $\log \Pi + \log H$ predicts well:** $R^2 = 0.722$ against a ceiling of 0.792, ahead of SNSP’s 0.652, with two fitted coefficients. Its earlier collapse was an artefact of Layer 2; its earlier brilliance was an artefact of Layer 1.
4. **Told which contingency list it is being assessed against — which needs the list, not its outcomes — it reaches $R^2 = 0.940$ on the RoCoF and 0.999 on the deviation,** against SNSP’s 0.652 and 0.748. The hazard aggregate of a new list is computable without a solve, so the model adapts to a change of list; SNSP structurally cannot. This is the strongest honest claim available, and Figure 1 is drawn on it.
5. **No family is separable from another here.** All the bootstrap intervals overlap. Parsimony is then the tie-breaker, and it favours the two-column model over twenty or thirty spatial columns.
6. The spectral clustering remains untested rather than refuted, for the reason in Section 5: at 88–95% system-wide content there is too little local variance for a spatial feature to resolve. It is nominally top of Table 2, but not separably so.

Caveats. With 60 held-out configurations, differences below about 0.02 in R^2 are not resolvable, and several comparisons above sit inside that. The `modal` family fits 31 columns on 180 training rows, a thin ratio even under ridge. Demand is constant across this ensemble, which flatters SNSP. The toy grid offers only about twenty-four possible disturbance locations (thirteen farms, five cities, six infeeds), so two sixty-event draws share twenty-one of them and differ mainly in severity and multiplicity rather than in place; a genuinely independent contingency list on a larger network would be a stronger test. Layer 3 also assumes the prior is correct — it tests generalisation across draws, not against a misspecified hazard model, which `risk_predictor.cross_hazard_model` probes separately and where the hazard term does lose to inertia alone.